

SANTA BARBARA 11 W, CA, USAWMO: **745050**Lat: **34.4141N**Lon: **119.8796W**Elev: **18**StdP: **14.69**Time Zone: **-8.00 (NAP)**Period: **08-19**WBAN: **53152****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	37.2	39.3	23.7	17.9	50.9	26.7	20.7	49.4	14.5	58.0	12.8	57.4	2.2	N/A	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	14.9	79.5	65.1	76.4	64.9	74.2	64.3	69.4	76.0	67.9	74.1	66.5	72.4	5.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.9	99.2	74.0	65.2	93.2	72.0	63.9	89.1	70.6	33.5	75.7	32.1	74.3	31.1	72.7	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	10.7	8.7	DB	33.4	88.3	2.1	4.5	31.9	91.5	30.6	94.1	29.4	96.7	27.9	99.9	(4)
(5)			WB	29.4	71.0	1.7	1.6	28.2	72.1	27.2	73.1	26.2	74.0	24.9	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	53.5	53.3	55.3	57.2	59.1	61.3	64.4	64.6	64.8	62.2	57.2	52.8	(6)	
	DBStd	5.71	4.04	3.92	4.02	3.66	3.02	2.35	3.19	3.24	3.99	4.25	3.88	4.03	(7)	
	HDD50	47	14	11	2	1	0	0	0	0	0	0	2	17	(8)	
	HDD65	2386	356	328	301	236	187	114	49	46	49	108	235	377	(9)	
	CDD50	3281	123	103	168	217	284	340	446	453	444	380	219	104	(10)	
	CDD65	142	0	0	1	2	5	4	30	34	42	23	1	0	(11)	
	CDH74	582	4	3	13	18	39	12	62	73	170	160	27	1	(12)	
	CDH80	94	0	0	1	2	8	0	14	4	36	26	3	0	(13)	
(14) Wind	WSAvg	3.7	3.3	3.8	4.0	4.3	4.2	3.7	3.8	3.5	3.4	3.4	3.4	3.5	(14)	
(15) Precipitation	PrecAvg	16.0	3.2	3.7	3.1	0.9	0.2	0.0	0.0	0.1	0.1	0.5	2.0	2.2	(15)	
	PrecMax	35.1	12.2	13.6	11.5	5.7	1.8	0.5	0.4	1.9	1.3	2.4	6.9	5.8	(16)	
	PrecMin	4.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	6.7	3.1	3.3	2.8	1.4	0.4	0.1	0.1	0.3	0.3	0.6	2.0	1.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.4	73.7	74.9	76.5	79.9	75.8	80.4	79.4	85.6	83.6	77.5	70.4	(19)	
		MCWB	53.2	56.4	58.0	55.9	57.4	62.2	68.6	68.9	62.7	60.7	59.6	56.4	(20)	
	2%	DB	69.0	68.0	69.1	70.6	71.5	71.1	74.7	75.9	78.2	78.3	72.9	67.1	(21)	
		MCWB	54.1	56.0	57.8	57.0	58.8	62.7	66.4	67.7	68.3	63.3	59.3	57.0	(22)	
	5%	DB	65.6	64.5	66.0	67.6	68.2	69.3	72.7	73.6	75.0	74.7	69.4	64.4	(23)	
		MCWB	55.1	56.1	57.2	57.2	58.8	62.0	65.7	66.3	66.4	63.6	58.5	56.1	(24)	
	10%	DB	63.0	61.9	63.5	65.1	66.1	67.8	70.7	71.7	72.7	72.0	67.2	62.3	(25)	
		MCWB	55.4	55.4	56.3	56.3	58.2	61.2	64.4	65.0	65.3	63.1	58.6	55.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	61.1	62.4	62.0	62.3	65.4	69.7	70.4	71.6	70.8	64.3	63.3	(27)	
		MCDB	64.6	64.9	69.8	69.1	69.6	71.9	77.7	78.0	78.2	75.9	70.8	66.6	(28)	
	2%	WB	59.0	59.4	60.2	60.2	60.9	64.0	67.6	68.3	69.7	68.3	62.2	60.5	(29)	
		MCDB	62.5	63.8	66.0	66.4	67.3	70.2	74.1	75.4	76.5	73.8	68.3	64.3	(30)	
	5%	WB	57.9	58.2	59.0	59.1	59.8	63.0	66.2	66.9	68.0	65.9	61.0	58.9	(31)	
		MCDB	61.6	62.4	64.0	64.8	65.8	68.7	72.3	73.4	73.9	72.3	66.9	62.6	(32)	
	10%	WB	56.8	57.0	57.5	57.9	59.0	61.9	65.1	65.7	66.5	64.2	59.7	57.1	(33)	
		MCDB	60.7	61.0	61.6	63.1	64.6	67.2	70.8	71.5	72.0	70.6	65.4	61.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.0	17.2	15.5	15.5	14.6	12.5	12.4	13.0	14.9	18.4	19.9	18.6	(35)	
		MCDBR	24.4	22.2	20.4	20.8	19.1	16.0	14.4	15.4	18.5	23.6	24.5	22.1	(36)	
	5% WB	MCWBR	15.8	14.7	13.0	11.9	10.6	9.2	7.6	8.5	10.4	14.2	17.0	16.2	(37)	
		MCWBR	17.4	17.3	16.6	16.8	15.5	14.4	13.8	14.9	16.0	19.6	21.6	17.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.317	0.326	0.362	0.378	0.364	0.393	0.372	0.362	0.350	0.308	0.311	(40)	
		taud	2.595	2.569	2.545	2.425	2.402	2.444	2.415	2.441	2.479	2.504	2.601	2.588	(41)	
	Ebn at Noon		290	297	302	293	288	291	281	286	285	280	286	280	(42)	
		Edh at Noon	24	28	31	36	38	36	37	35	32	29	24	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		974	1249	1685	2100	2313	2397	2316	2168	1859	1428	1098	882	(44)	
	RadStd		74	130	109	120	154	111	93	84	78	106	60	71	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	+1.10	N/A	-2.92	+1.22	N/A	N/A	N/A	-229	+396	+149

Nomenclature: See separate page